

OS Practical 11

```
#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define ITERATIONS 1000000

long long counter = 0;

void *increment(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {
        counter++;
    }

    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    for (int i = 0; i < THREADS; i++) {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    for (int i = 0; i < THREADS; i++) {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter = %d\n", THREADS * ITERATIONS);
    printf("Actual counter    = %lld\n", counter);

    return 0;
}
```

```
nishanth@LAPTOP-6Q3ON07H:~/q11$ gcc race_counter.c -o race_
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1482562
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1182614
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1489107
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1443161
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1446098
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1487099
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1423218
```

```

#include <stdio.h>
#include <pthread.h>

#define THREADS 4
#define ITERATIONS 1000000

long long counter = 0;

pthread_mutex_t lock;

void *increment(void *arg) {
    for (int i = 0; i < ITERATIONS; i++) {

        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main() {
    pthread_t threads[THREADS];

    pthread_mutex_init(&lock, NULL);

    for (int i = 0; i < THREADS; i++) {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    for (int i = 0; i < THREADS; i++) {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter = %d\n", THREADS * ITERATIONS);
    printf("Actual counter    = %lld\n", counter);

    pthread_mutex_destroy(&lock);

    return 0;
}

```

```

nishanth@LAPTOP-6Q3ON07H:~/q11$ nano mutex_counter.c
nishanth@LAPTOP-6Q3ON07H:~/q11$ gcc mutex_counter.c -o mutex_counter -pthread
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000
nishanth@LAPTOP-6Q3ON07H:~/q11$ time ./race_counter
Expected counter = 4000000
Actual counter   = 1441319

real    0m0.033s
user    0m0.103s
sys     0m0.008s
nishanth@LAPTOP-6Q3ON07H:~/q11$ time ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000

real    0m0.253s
user    0m0.301s
sys     0m0.486s

```

```

nishanth@LAPTOP-6Q3ON07H:~/q11$ gcc race_counter.c -o race_counter -pthread
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1349027
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1325902
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./race_counter
Expected counter = 4000000
Actual counter   = 1504545
nishanth@LAPTOP-6Q3ON07H:~/q11$ gcc mutex_counter.c -o mutex_counter -pthread
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000
nishanth@LAPTOP-6Q3ON07H:~/q11$ ./mutex_counter
Expected counter = 4000000
Actual counter   = 4000000

```

